PROBE

mechanistic normal tissue complication
PRobability mOdeling for particle BEam radiation

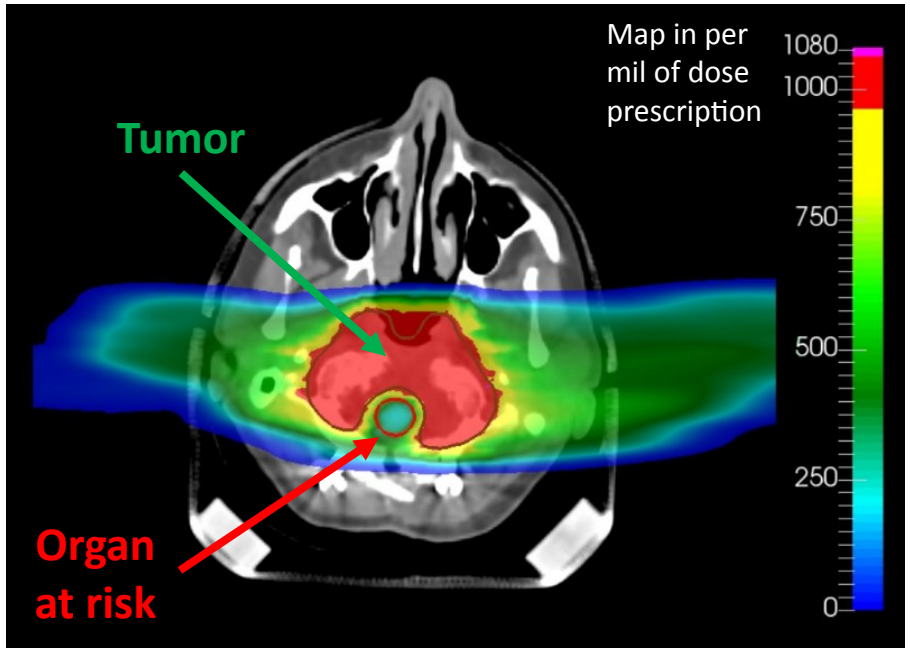
Marco Battestini, Ph.D.

INFN CSN5 “Grant Giovani” proposal, Rome, 01/12/2025

Particle Therapy (PT)

General goals of RadioTherapy (RT):

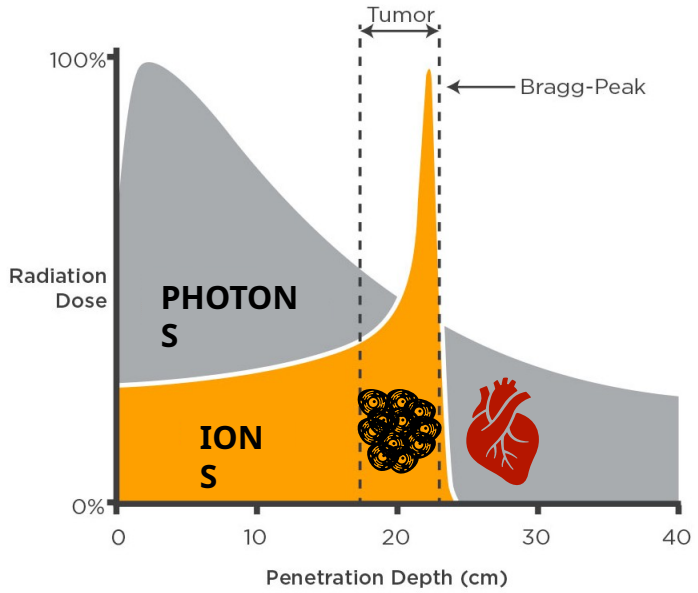
1) **Controlling** the **tumor**
2) **Sparing** as much as possible **healthy tissues**, in order to **reduce** normal tissue **toxicity**

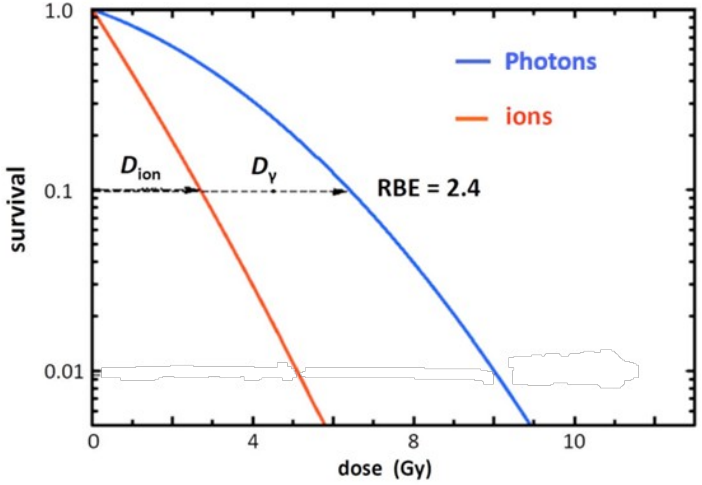


M. Battestini et al., Front. Oncol. (2022)

INFN Istituto Nazionale di Fisica Nucleare — MoVe IT

PT: physical and biological ($Z > 1$) advantages





U. Amaldi et al., CERN Yellow Reports (2019)

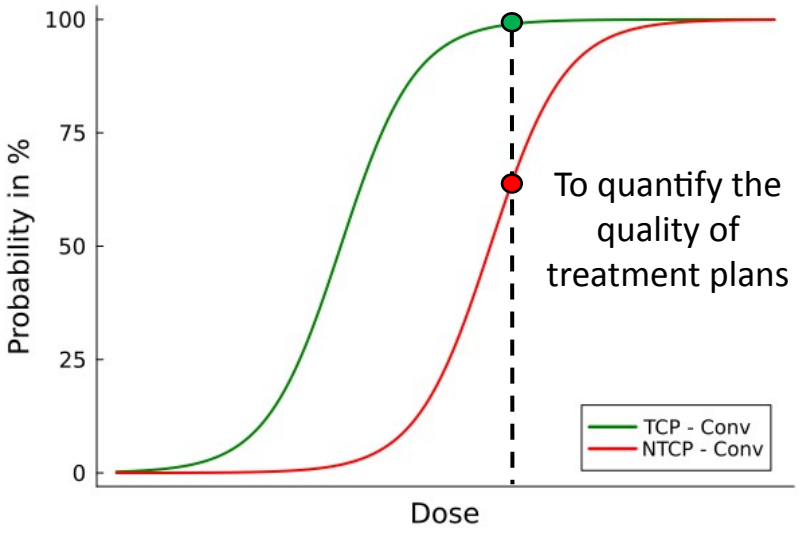
Relative Biological Effectiveness ⟶ clinics ⟶ from *in-vitro* data

$$RBE = \frac{D_{photon}}{D_{ion}}\Big|_{\text{isoeffect}}$$

PROBE — mechanistic normal PRobability mOdeling — tissue complication for particle BEam radiation

NTCP models in literature for standard RadioTherapy (RT)

Tumor Control Probability (TCP)
Normal Tissue Complication Probability (NTCP)



To quantify the quality of treatment plans

Phenomenological models
e.g. Lyman-Kutcher-Burman (LKB) model

NTCP

Statistical models
e.g. uni- and multi-variate (MVA) logistic analyses

Mechanistic models
e.g. **Relative Seriality model**

Particle Therapy (PT)

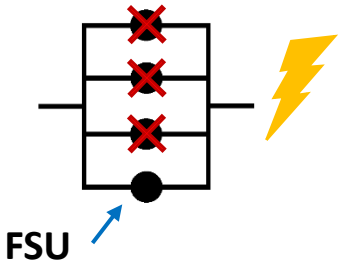
PROBE
mechanistic normal PRobability mOdeling
tissue complication for particle BEam radiation

Lacking a solid foundation in **radiation biophysics**

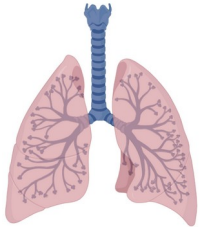
Tissue architecture and volume effect

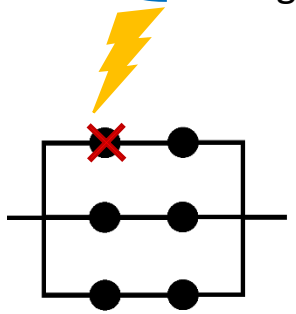
Tissue architecture → Functional sub-units (**FSU**s) arrangement

Parallel organ



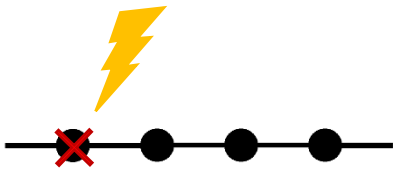
- Any FSU remaining preserves organ function
- Toxicity induction linked to **mean dose**
- **Large volume effect**
- e.g. lung, parotid gland





→ More realistic scenario

Serial organ



- Inactivation of any FSU leads to complication
- Toxicity induction linked to **maximum dose**
- **Small volume effect**
- e.g. spinal cord, oesophagus



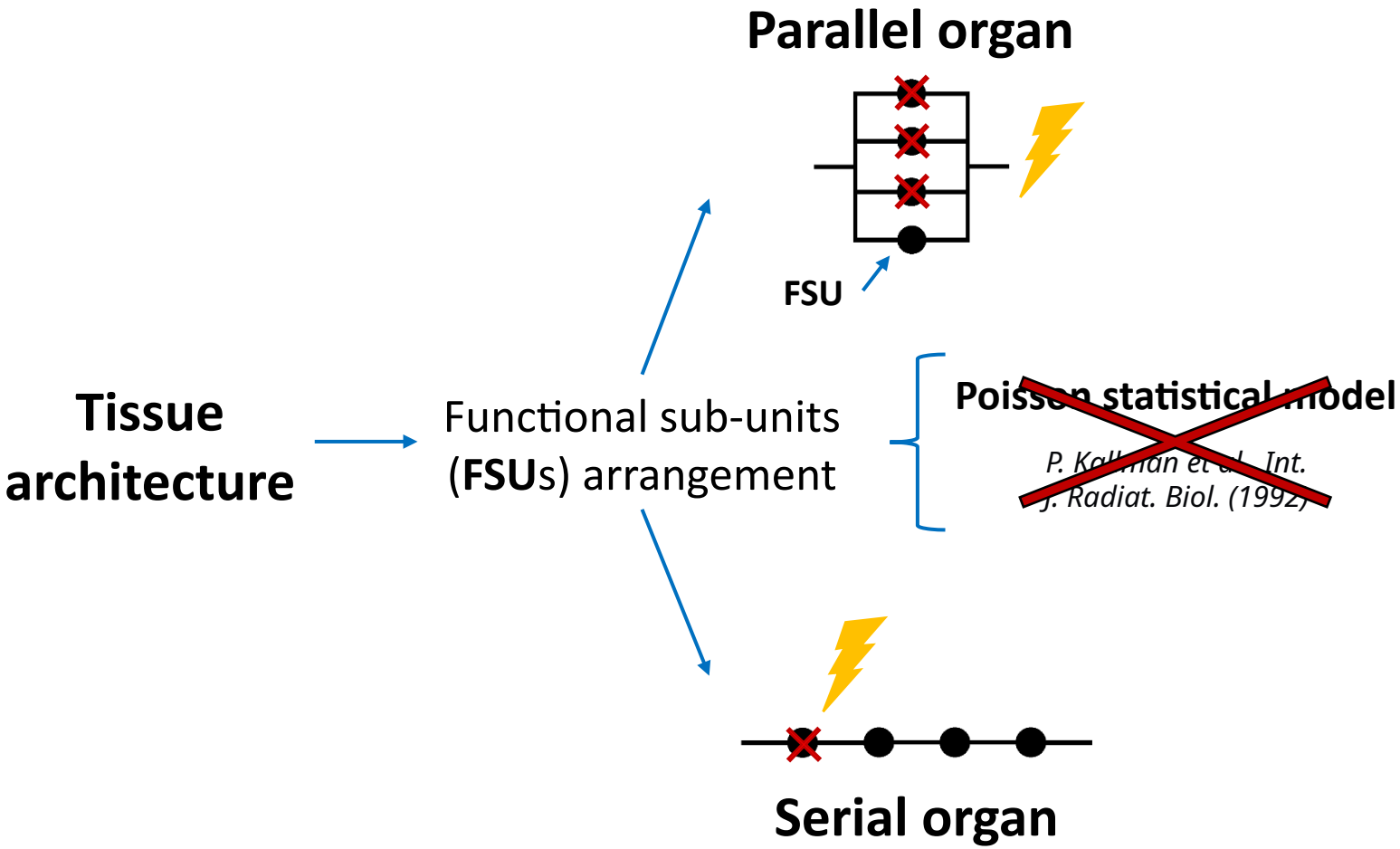
Volume effect

How an organ responds to a damage depending on its tissue architecture

M. Battestini et al., Front. Oncol. (2022)

P. Kallman et al., Int. J. Radiat. Biol. (1992)

Mechanistic NTCP: the Relative Seriality model



PROBE
mechanistic normal PRobability mOdeling
tissue complication for particle BEam radiation

Physical and biological **stochasticities** and bio-chemical environmental **heterogeneities**

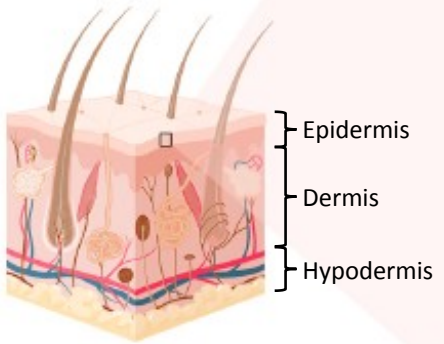
Radiation-induced skin toxicity

PROBE
mechanistic normal tissue complication
PRobability mOdeling for particle BEam radiation

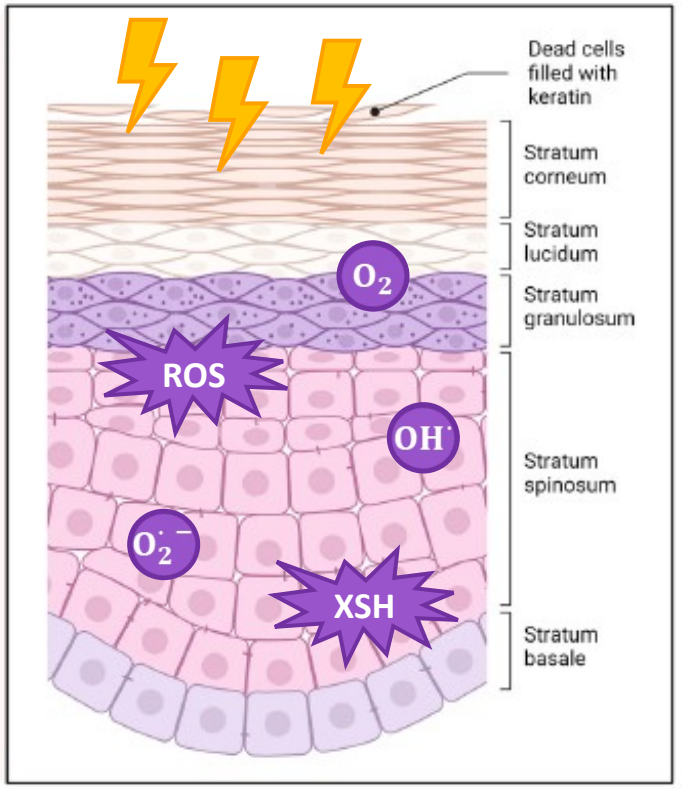
Skin toxicity

Common in all radiation treatments

NTCP models in literature



Rübe et al., Int. J. Mol. Sci. (2024)



Y. Shindo et al., J. Invest. Dermatol. (1994)

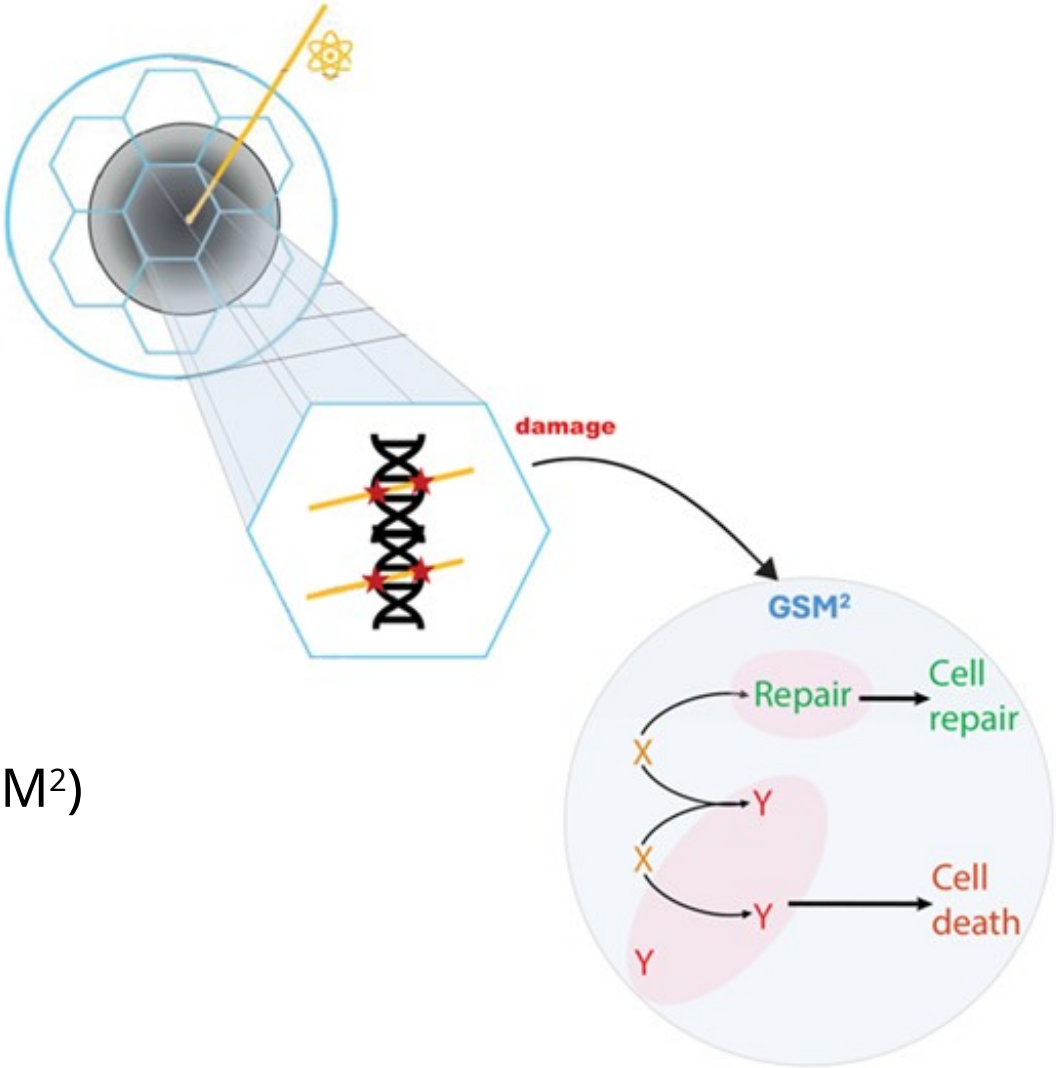
N. Chettouh-Hammas et al., Oxid. Med. Cell. Longev. (2024)

Complex **multi-layered architecture** with specific bio-chemical properties and functions

Radiation toxicity influenced by **radiation chemistry** (free radicals, ROS, XSH, O_2)

Modeling starting point

The Generalized Stochastic Microdosimetric Model (GSM^2)



The Generalized Stochastic Microdosimetric Model (GSM²)

1) DNA damage formation

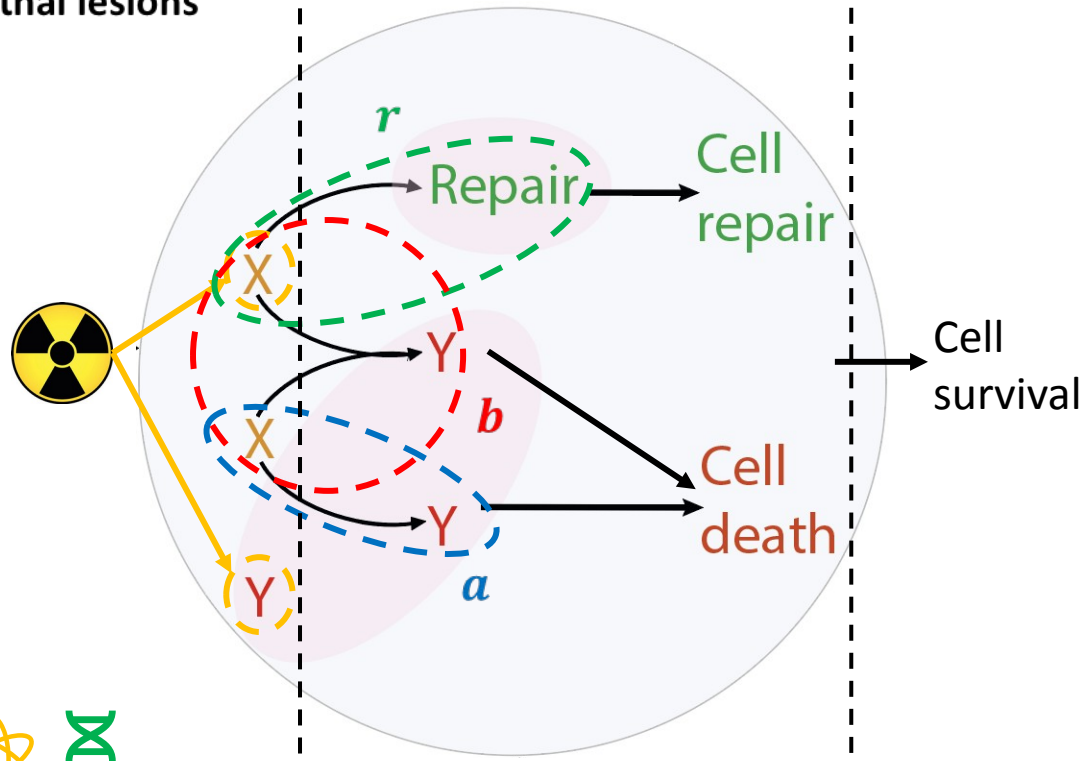
X: number of **sub-lethal lesions**

Y: number of **lethal lesions**

2) DNA damage evolution

r, a, b depend on cell type

3) Biological endpoint



GSM²: a fully probabilistic radiation biophysical model → Different levels of **spatio-temporal stochasticity** in physics and biology

Development and validation

INFN Istituto Nazionale di Fisica Nucleare — Microbe_IT — MUSICA

F. G. Cordoni et al., Phys. Rev. E (2021)

F. G. Cordoni et al., Rad. Res. (2022)

F. G. Cordoni et al., Int. J. Radiat. Biol. (2023)

G. Bordieri,..., M. Battestini, et al., Phys. Med. Biol. (2024)

Extensions

INFN Istituto Nazionale di Fisica Nucleare — FLASH Radiotherapy with high Dose-rate particle beAms — MIRO Minibeam Radiotherapy

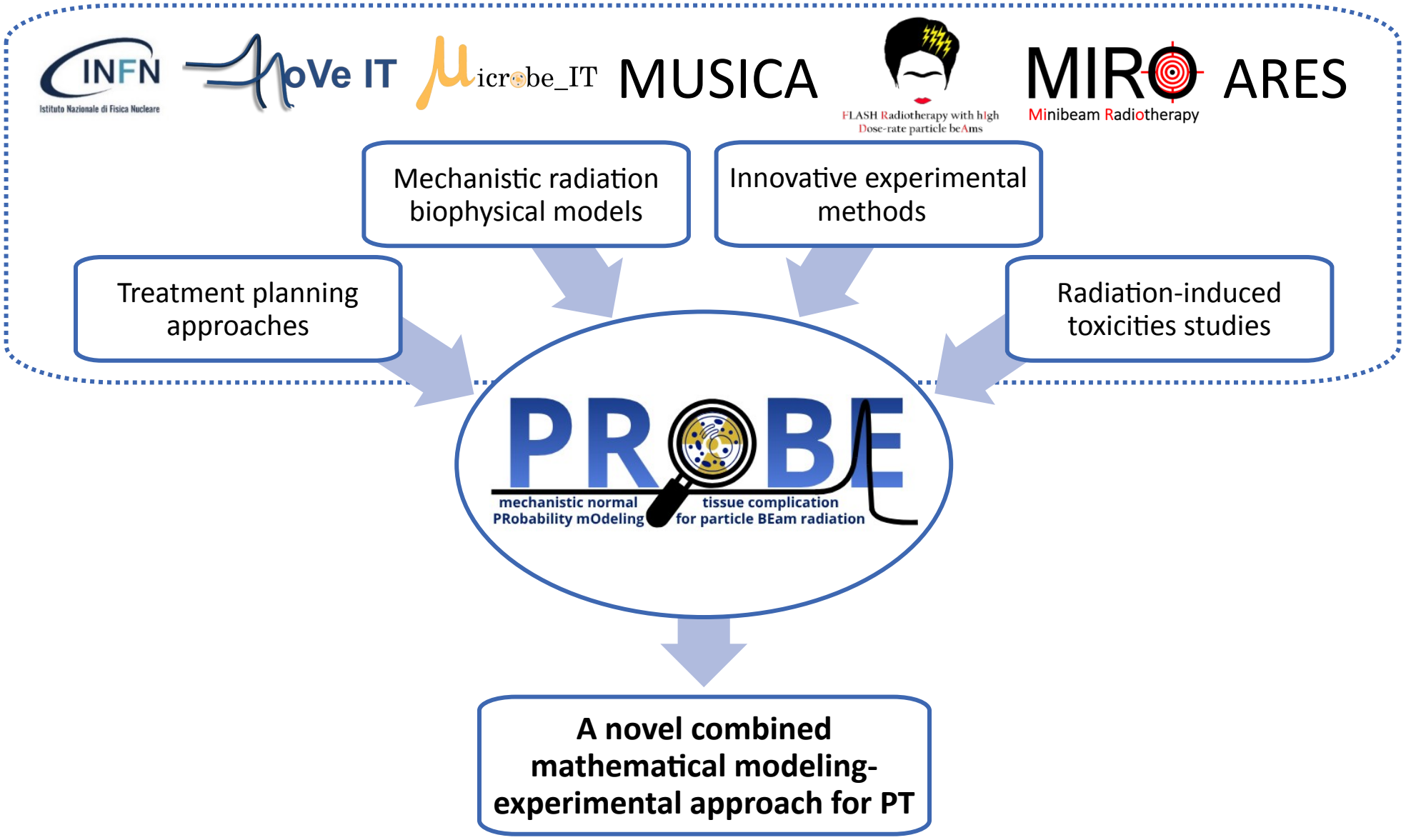
M. Battestini et al., Front. Phys. (2023)

M. Battestini et al., Radiother. Oncol. (2025)

The PROBE project

mechanistic normal tissue complication
PRobability mOdeling for particle BEam radiation

The main goal of the project



Workpackages organization

PROBE
mechanistic normal tissue complication
PRobability mOdeling for particle BEam radiation

WP1: NTCP modeling for particle therapy

WP1.1
- Development of a general mechanistic-driven NTCP model
- Systematic analysis of model parameters

WP1.2
- Applications of the NTCP model to *in-vivo* experimental data from literature

WP1.3
- Model extension towards clinically relevant predictions for simple geometries

TIFPA · UNIVERSITÀ DI TRENTO · Azienda Provinciale per i Servizi Sanitari Provincia Autonoma di Trento · PROTONTERAPIA TRENTO

INFN PA · UNIVERSITÀ DI PAVIA · INFN NAPOLI · Università degli Studi di Napoli Parthenope

WP2.1
- Irradiation of multi-layer tissues with X-ray and proton beams
- Bio-chemical and histological analyses

WP2.2
- Development of a specific mechanistic-driven model for skin lesions

WP2.3
- Calibration of bio-chemical model parameters on in-house skin data

WP2: Skin toxicity modeling and proton beam experiments

The workflow of the GSM²-based NTCP model

General model development

WP1.1

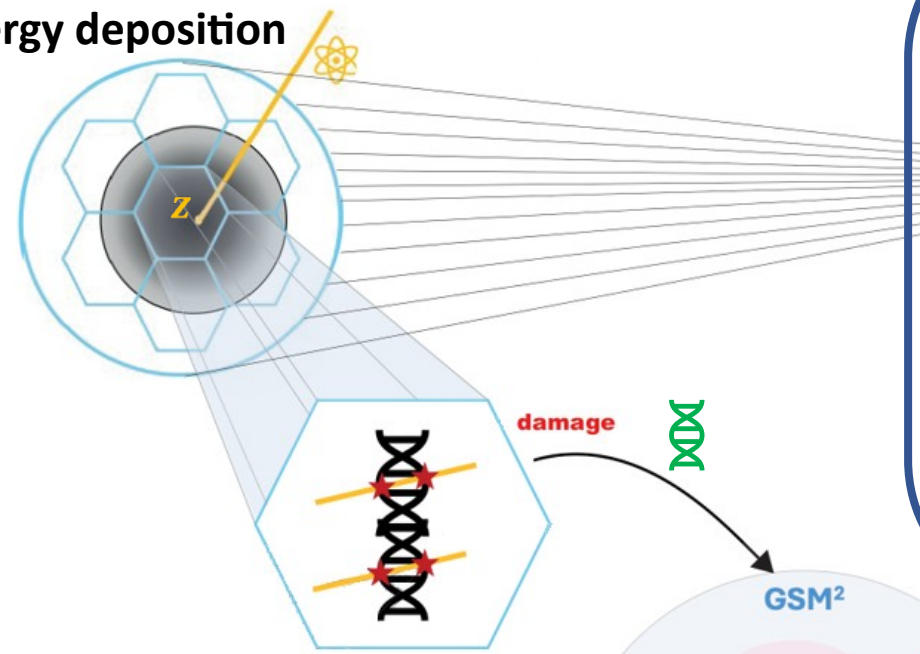
General model validation

WP1.2

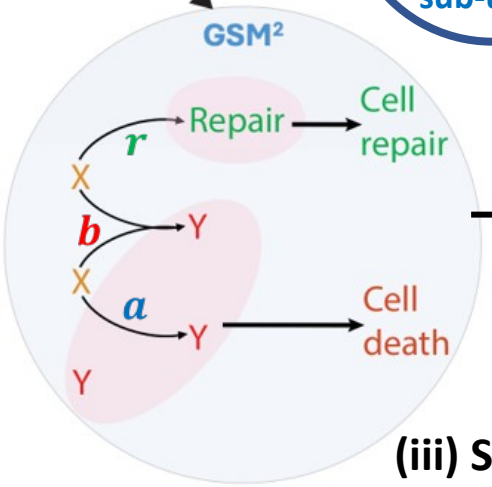
Model extension to clinic

WP1.3

(i) Energy deposition



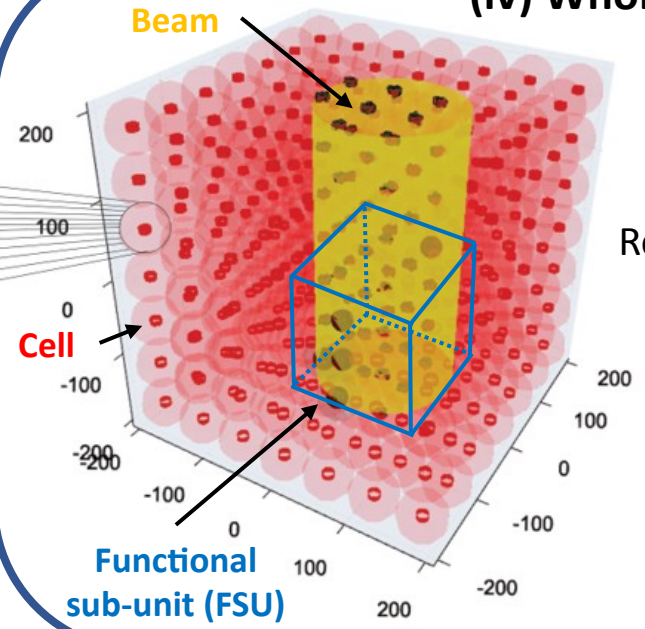
(ii) DNA damage formation



Cell survival

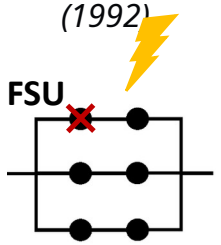
(iii) Single-cell response

(iv) Whole organ effect



NTCP/TCP
Relative Seriality model

P. Kallman et al., Int. J. Radiat. Biol. (1992)



Main advantage: inclusion of intra- and inter-cellular effects

From single cell description to multi-cellular response

General model development

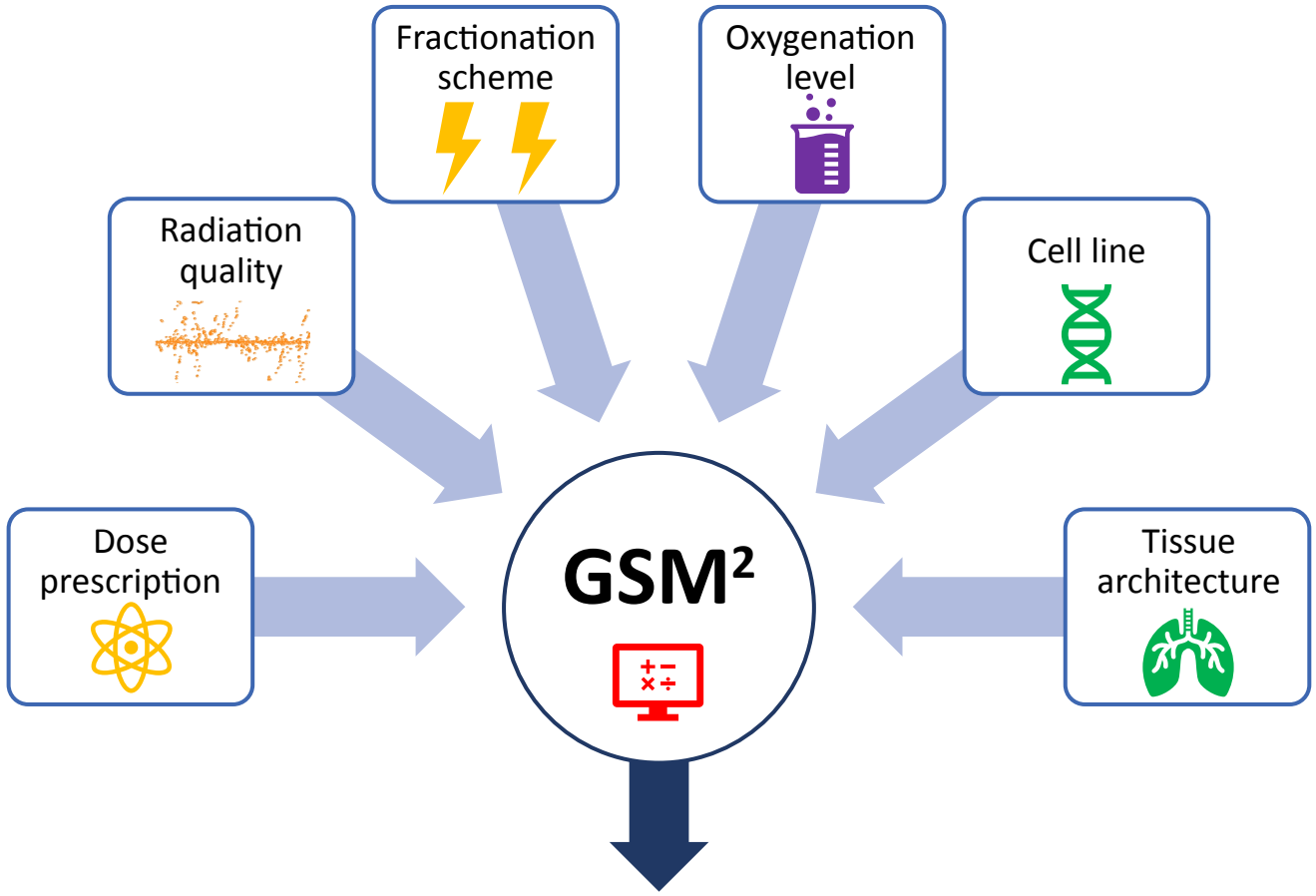
WP1.1

General model validation

WP1.2

Model extension to clinic

WP1.3



- Impact of irradiation methods and environmental conditions
- Impact of stochastic effects and heterogeneities
- GSM²-based NTCP model VS NTCP models from literature

TIFPA

UNIVERSITÀ DI TRENTO

Comparison of model predictions with experimental data

General model development

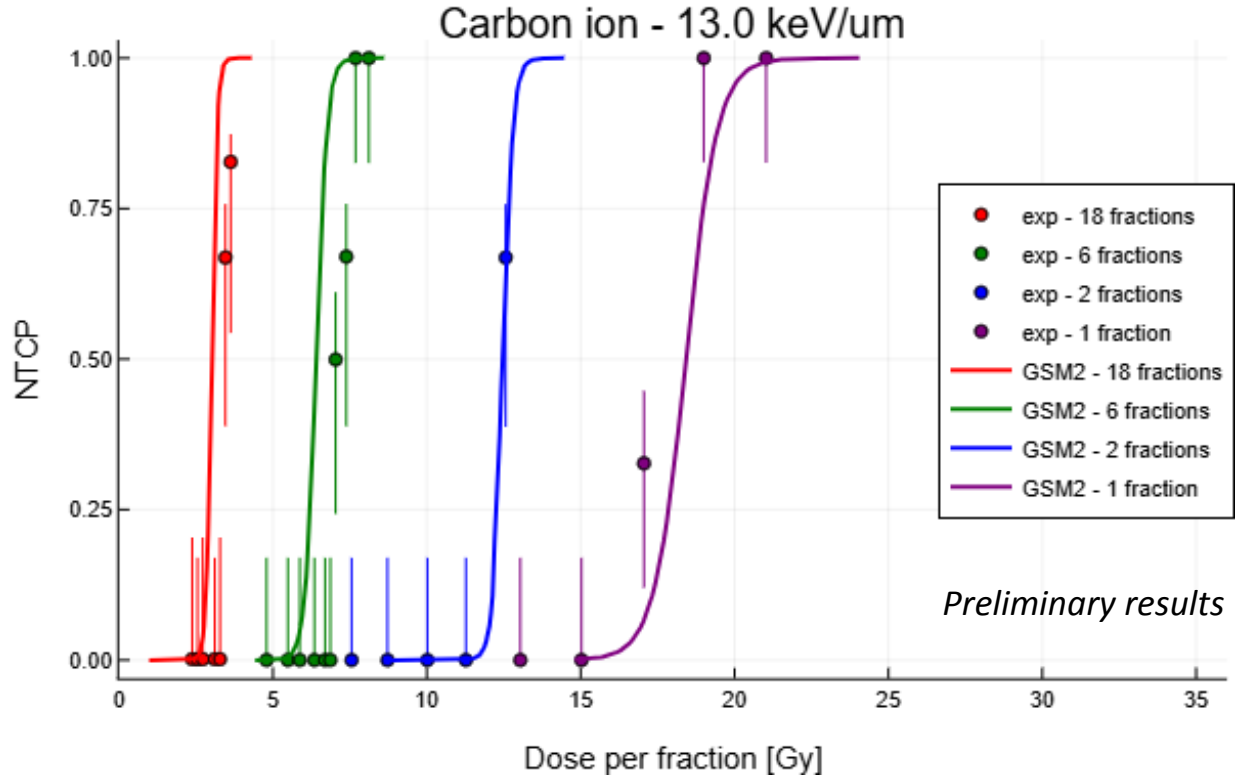
WP1.1

General model validation

WP1.2

Model extension to clinic

WP1.3



Prof. C. P. Karger (see letter)

dkfz. GERMAN CANCER RESEARCH CENTER IN THE HELMHOLTZ ASSOCIATION

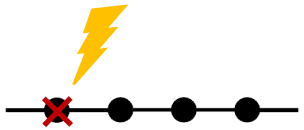
Research for a Life without Cancer

- Rat spinal cord
- Conv irradiation
- ^{12}C, ^{4}He, p
- 18, 6, 2, 1 fractions



Exp: M. Saager et al., Radiother. Oncol. (2018), L. Hintz et al., Radiother. Oncol. (2022), C. P. Karger et al., Int. J. Radiation Oncology Biol. Phys. (2006)

Assumptions: fully serial organ, single cell line, uniform oxygenation



- Complex parallel organs (e.g. lung) predictions
- Possible interplays among radiation quality, fractionation scheme and biological endpoint
- Tumor response predictions



TIFPA

UNIVERSITÀ DI TRENTO

Model extension for clinically relevant applications

General model development

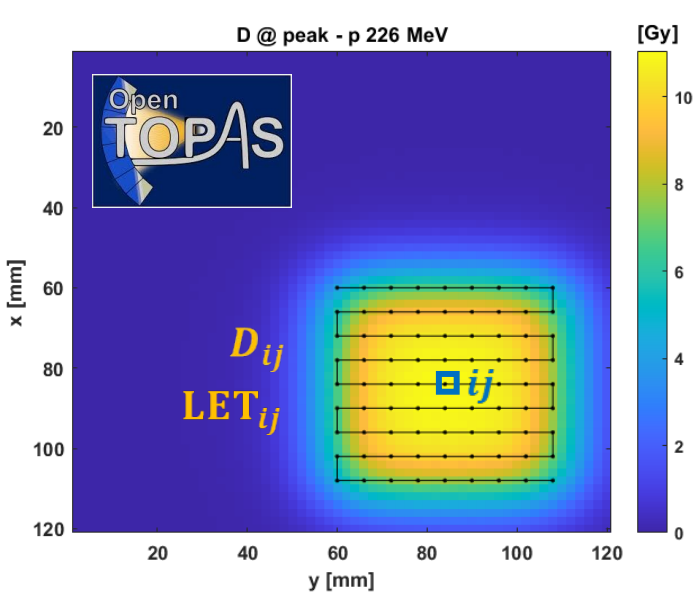
WP1.1

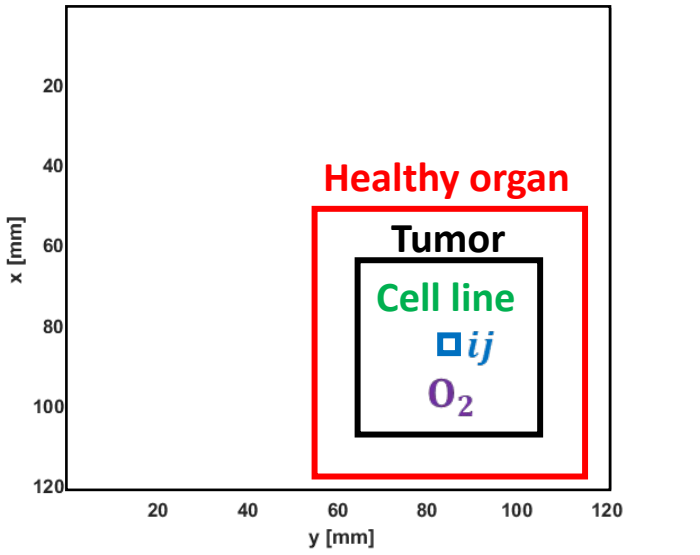
General model validation

WP1.2

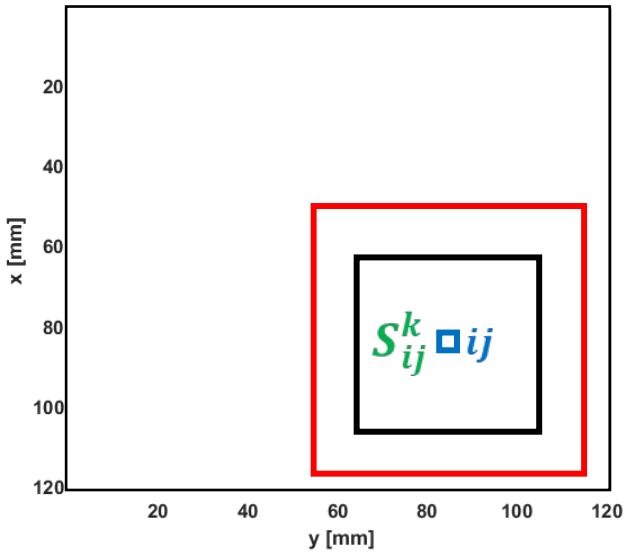
Model extension to clinic

WP1.3





GSM²



TCP/ NTCP

Tumor/organ response

Prof. F. Lohr (see

Azienda Provinciale per i Servizi Sanitari
Provincia Autonoma di Trento

PROTONTERAPIA TRENTO

Possible spillovers

Biophysics-driven plan optimization methods

TIFPA

UNIVERSITÀ DI TRENTO

Irradiation of biological multi-layer systems and analyses

Irradiation of 3D skin tissue

WP2.1

Skin model development

WP2.2

Skin model calibration

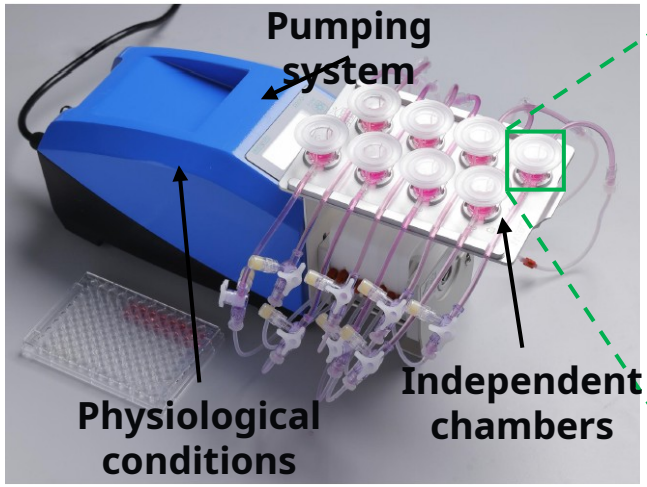
WP2.3

INFN Istituto Nazionale di Fisica Nucleare

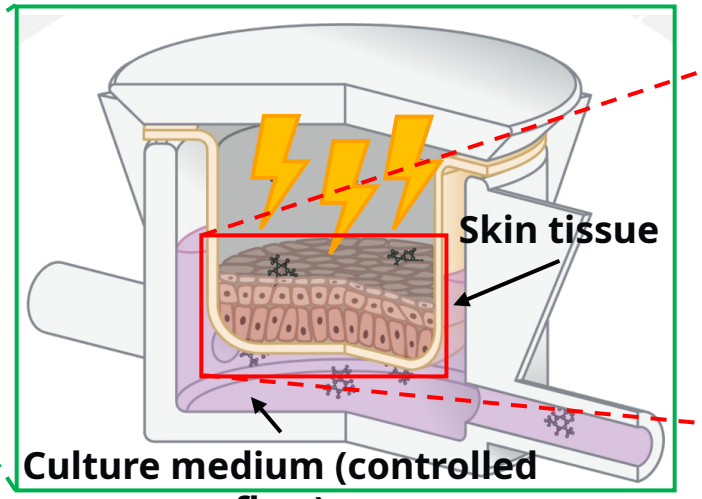
ARES

REACT4LIFE mirroring human complexity

Microfluidic bioreactor



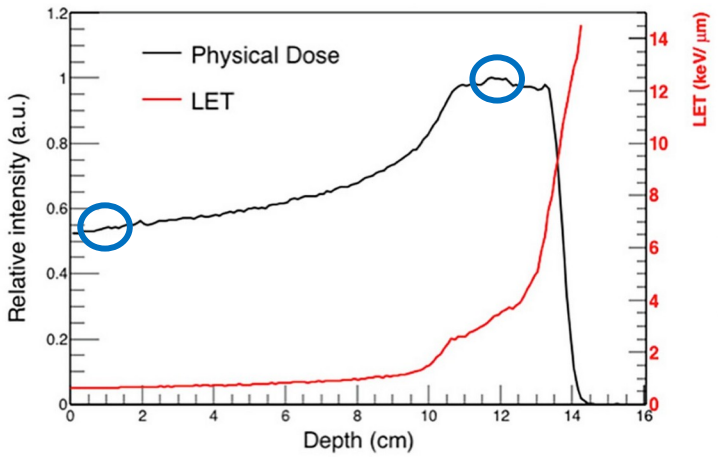
Skin-On-Chip



Multi-layer skin tissue



Experimental proton beam at Trento



F. Tommasino et al., Phys. Medica (2019)

Physical properties

Radiation quality

Fractionation scheme

- Number of fractions
- Dose per fraction
- Total dose

Chemical and biological analyses

- DNA damage
- Surviving fraction
- Histological analyses
- Oxidative stress
- Cytokines release

P. Simoniello et al., Front. Oncol. (2016)

TIFPA Istituto Nazionale di Fisica Nucleare

UNIVERSITÀ DI TRENTO

INFN PA VI

INFN NAP OLI

A specific mechanistic-driven NTCP model for skin toxicity

Irradiation of 3D skin tissue

WP2.1

Skin model development

WP2.2

Skin model calibration

WP2.3

INFN Istituto Nazionale di Fisica Nucleare

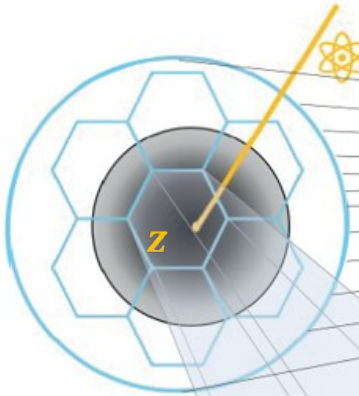
FLASH Radiotherapy with high Dose-rate particle beAms

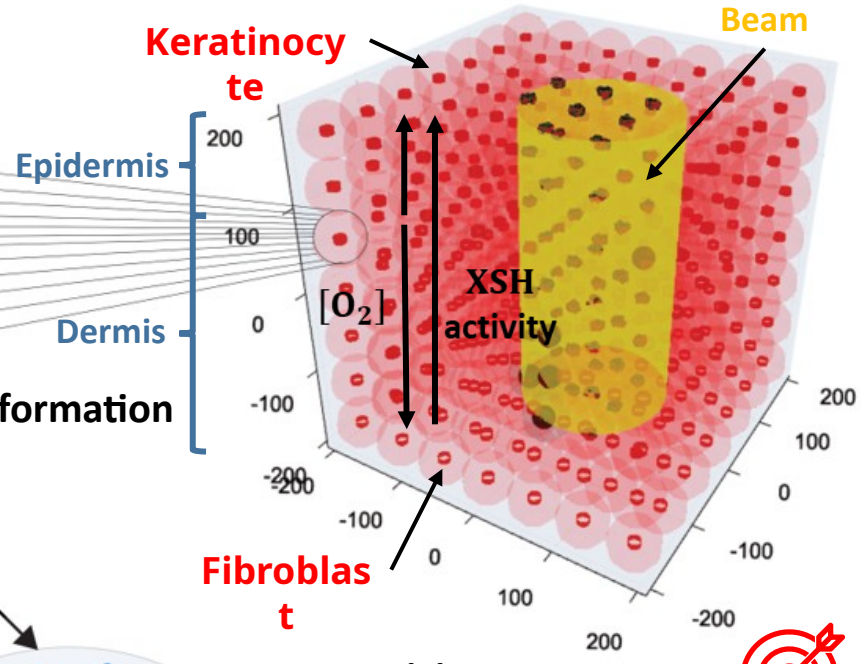
MS-GSM²

M. Battestini et al., Front. Phys. (2023)

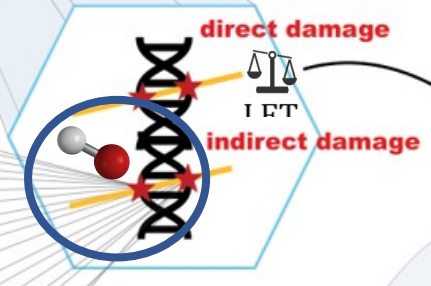
M. Battestini et al., Radiother. Oncol. (2025)

(i) Energy deposition

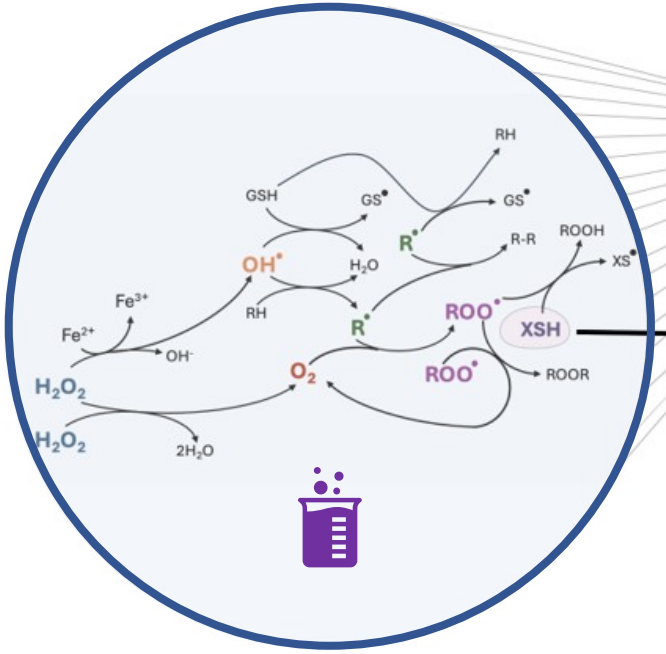




(iii) DNA damage formation



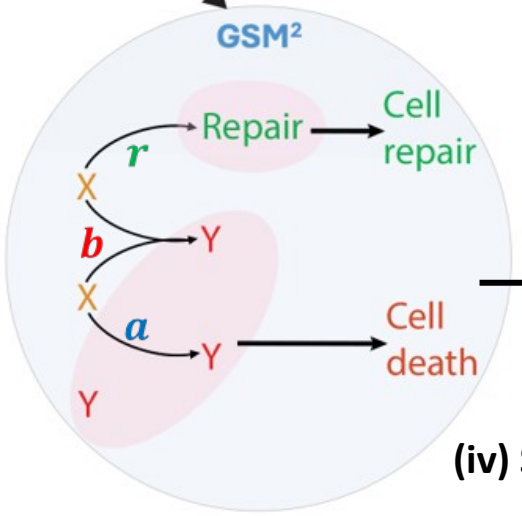
(ii) Chemistry evolution



Redox enviroment

↓

Antioxidant defense



Cell survival

(iv) Single-cell response

(v) Skin response

Bio-chemical model parameters calibration for skin lesions

Irradiation of 3D skin tissue

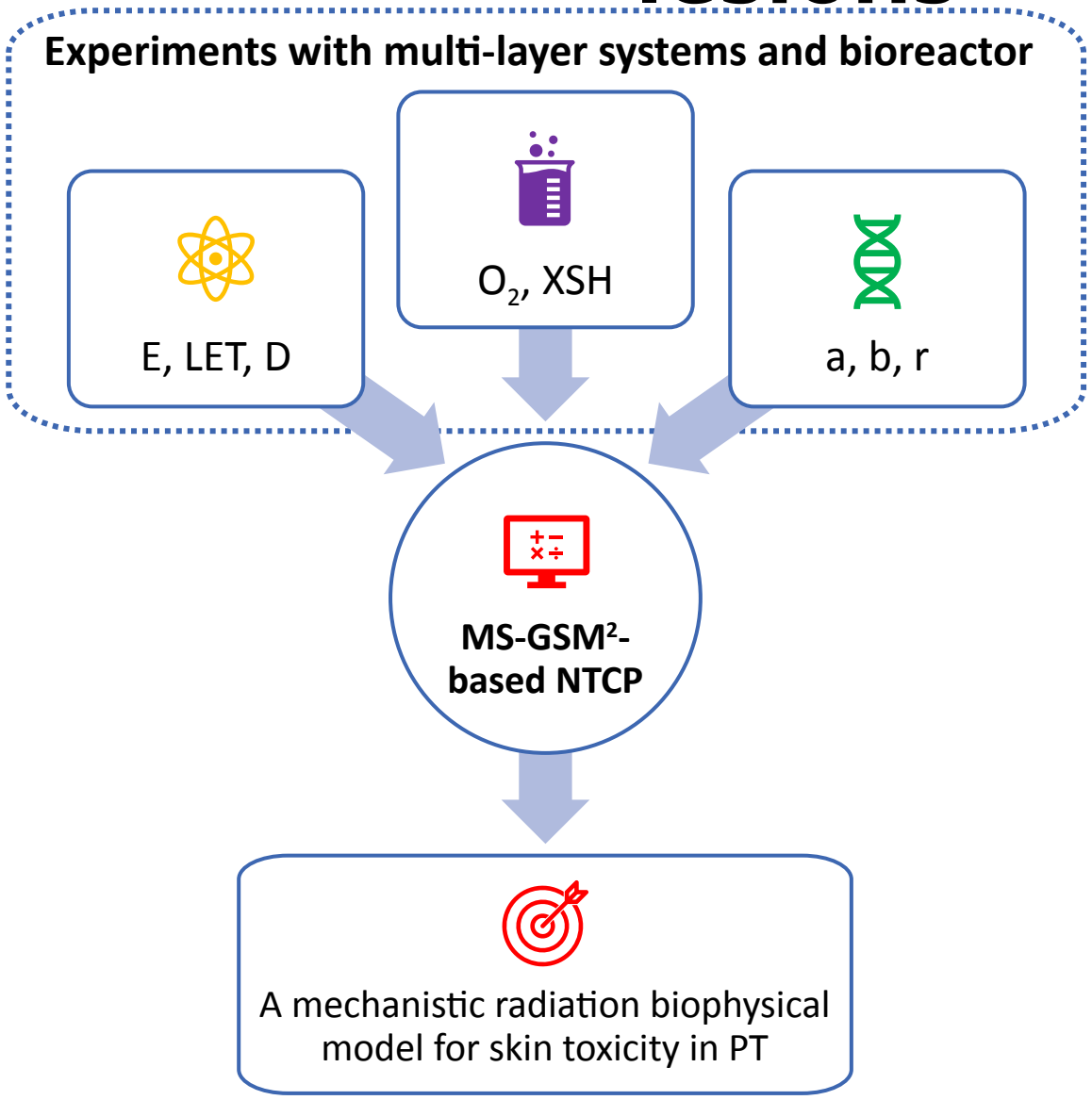
WP2.1

Skin model development

WP2.2

Skin model calibration

WP2.3



Possible spillovers

An integrated modeling-experimental approach to assess skin toxicity with innovative irradiation techniques

Spatially fractionated RT

INFN Istituto Nazionale di Fisica Nucleare

MIRO Minibeam Radiotherapy

FLASH RT

INFN Istituto Nazionale di Fisica Nucleare

FLASH Radiotherapy with hIgh Dose-rate particle beAms

M. Battestini et al., Front. Phys. (2023)

M. Battestini et al., Radiother. Oncol. (2025)

TIFPA

UNIVERSITÀ DI TRENTO

Gantt chart and milestones

First year	M. 1	M. 2	M. 3	M. 4	M. 5	M. 6	M. 7	M. 8	M. 9	M. 10	M. 11	M. 12
Literature review on NTCP models												
Development of the general framework of the novel NTCP/TCP model												
In-silico analysis of the impact of various parameters on NTCP/TCP predictions						1.1						
Literature review on in-vivo experiments and their rigorous selection												
Applications of the NTCP model to parallel organs from experimental data in the literature												1.2
Literature review on radiation-induced skin toxicities												
X-ray irradiation tests on the multi-layered biological system												
Initial NTCP model extension for skin injuries												
Proton irradiations of the multi-layered biological system - first part												

WP1.1: Development of a general mechanistic-driven NTCP model and systematic analysis of model parameters
WP1.2: Applications of the NTCP model to *in-vivo* experimental data from literature
WP1.3: Model extension towards clinically relevant predictions for simple geometries

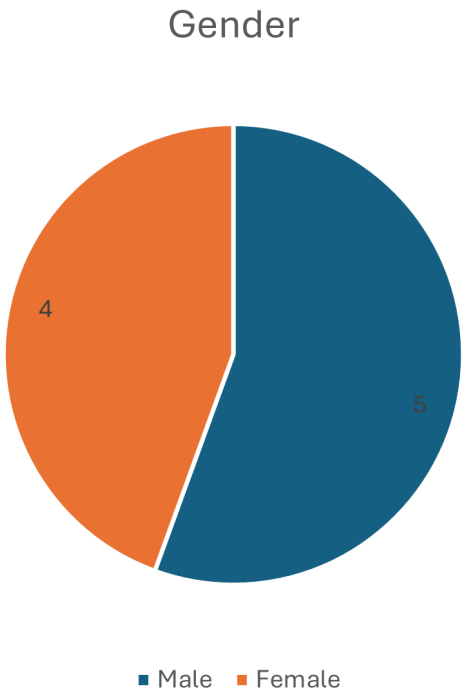
WP2.1: Proton irradiation of biological multi-layer systems with proton beams and bio-chemical analyses
WP2.2: Development of a specific mechanistic-driven model for skin lesions
WP2.3: Calibrations of bio-chemical model parameters on in-house experiments

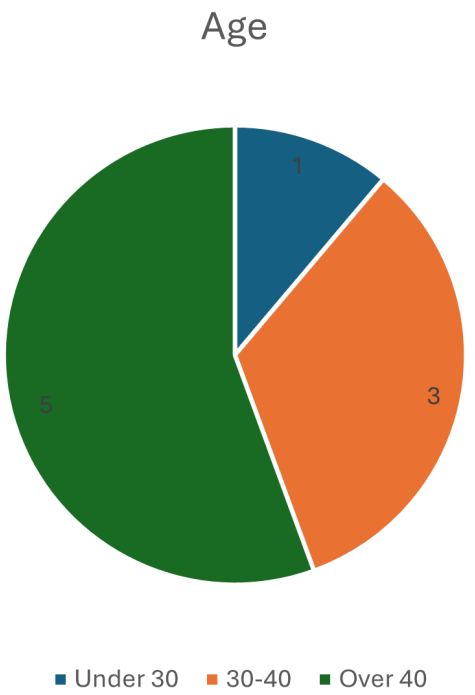
Second year	M. 1	M. 2	M. 3	M. 4	M. 5	M. 6	M. 7	M. 8	M. 9	M. 10	M. 11	M. 12
Development of the model extension for clinically relevant investigations												
Applications of the NTCP model to simple target geometries									1.3			
Proton irradiations of the multi-layered biological system - second part												
Chemical and biological analysis of irradiated biological samples							2.1					
Development of the specific NTCP model for skin lesions									2.2			
Applications of the specific NTCP model on in-house data for parameters calibration												2.3

Organization

Name	Position	Age	Unit	FTE
Marco Battestini (a, b)	PostDoc Researcher	29	INFN-TIFPA/UNITN	0.8
Emanuele Scifoni (c)	Senior Researcher	54	INFN-TIFPA	0.2
Francesco G. Cordoni	Associate Professor	37	INFN-TIFPA/UNITN	0.1
Francesco Tommasino	Associate Professor	40	INFN-TIFPA/UNITN	0.1
Alessandra Bisio	Associate Professor	49	INFN-TIFPA/UNITN	0.1
Silva Bortolussi (b)	Associate Professor	47	INFN-PV/UNIPV	0.1
Federica Riva	Associate Professor	52	INFN-PV/UNIPV	0.2
Ricardo Luis Ramos	RTT	40	INFN-PV/UNIPV	0.1
Palma Simoniello (b)	Associate Professor	53	INFN-NA/ UNIPARTHENOPE	0.3
			Total	**2.0**

Involved personnel. (a) Principal investigator, (b) coordinator of the research unit, (c) coordinator with a permanent contract.



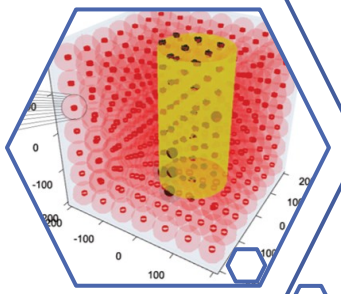


Costs

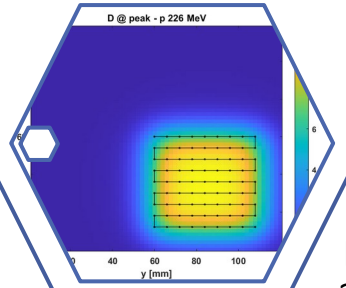
Richieste primo anno		
Missioni	Missioni per irraggiamento a Trento dai partners	3k €
Consumabili	Acquisto campioni biologici e kit analisi	15k €
Inventariabili	Acquisto 2 bioreattori	25k €
Servizi	Costo 2 turni fascio TPTC (Sub Judice approvazione PAC)	5k €
Risorse computazionali	Costo CPU (250k core hour) e storage (5TB-N)	2k €
	Totale primo anno (con IVA)	50k €
Richieste secondo anno		
Missioni	Missioni per irraggiamento a Trento dai partners	3k €
Consumabili	Acquisto campioni biologici e kit analisi	12k €
Servizi	Costo 3 turni fascio TPTC (Sub Judice approvazione PAC)	7.5k €
Risorse computazionali	Costo CPU (250k core hour)	1.5k €
	Totale secondo anno (con IVA)	24k €
Totale (con IVA)		74k €

PROBE

mechanistic normal tissue complication
PRobability mOdeling for particle BEam radiation



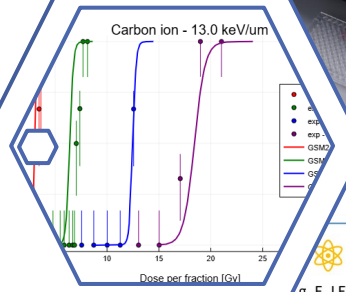
Model extension to clinic



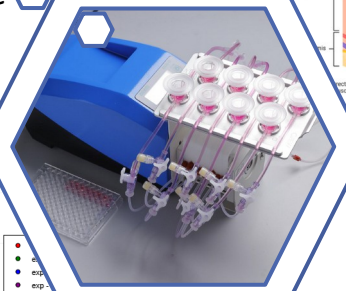
General model validation

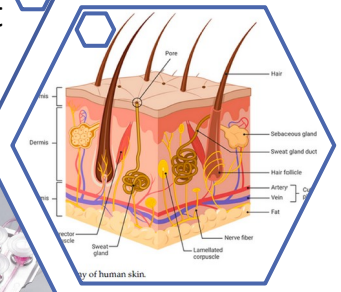
General model development

Irradiation of 3D skin tissue and analyses

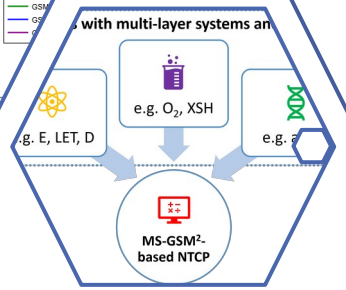


Skin model development





Skin model calibration



Backup slides

Risk assessment

Risk	Impact	Solution/mitigation
Delays in procuring the bioreactor and/or its malfunction	Delays in irradiation testing and/or a decreased number of measurement campaigns	We will focus on further model development and its in-silico applications (e.g. additional clinically relevant developments, model extensions for specific PT settings) and/or we will irradiate the multi-cellular system without the bioreactor
Malfunctions in the multi-cellular system and/or in its irradiation	Alternative and/or decreased number of measurement campaigns	We will assess irradiating simpler systems for calibration (e.g., 2D cell cultures) and/or alternative surrogate systems such as spheroids for chemical analyses
Delays in allocation of beamtime slots	Decreased number of measurement campaigns	We will consider involving additional centers for irradiation currently not foreseen in the original project framework (e.g. CNAO)
Malfunctions in the biological and/or chemical analyses of the irradiated tissues	Alternative and/or reduced number of analyses	We will carry out a detailed analysis of the available assays and/or we will assess the feasibility of additional analyses and/or we will focus on further model development and its in-silico applications (e.g. additional clinically relevant developments, model extensions for specific PT settings)
Delays and/or issues in the development and/or calibration of the specific NTCP framework for skin toxicity	Reduced number of biological endpoints available for analysis and incomplete model calibration	We will focus on the study of the available endpoints and/or we will calibrate the model directly on in-vivo data from literature and/or we will focus on further model development and its in-silico applications (e.g. additional clinically relevant developments, model extensions for specific PT settings)

The mathematical workflow of the proposed NTCP model

General model development

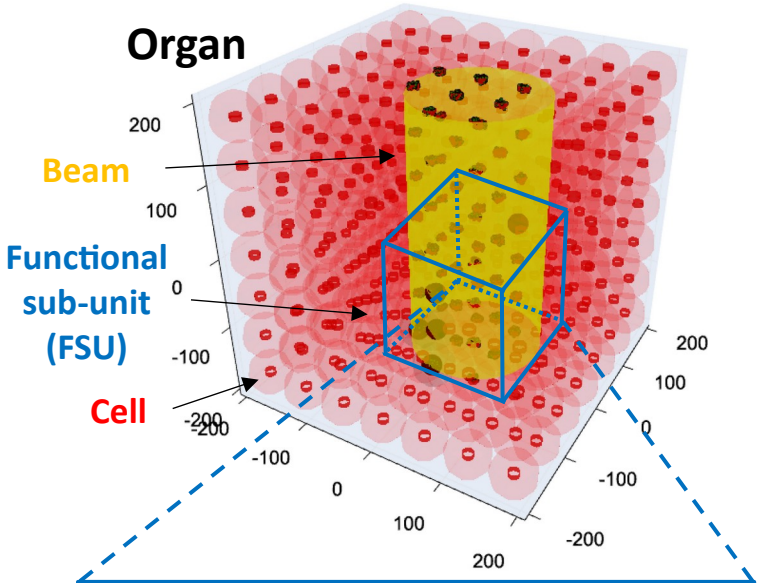
WP1.1

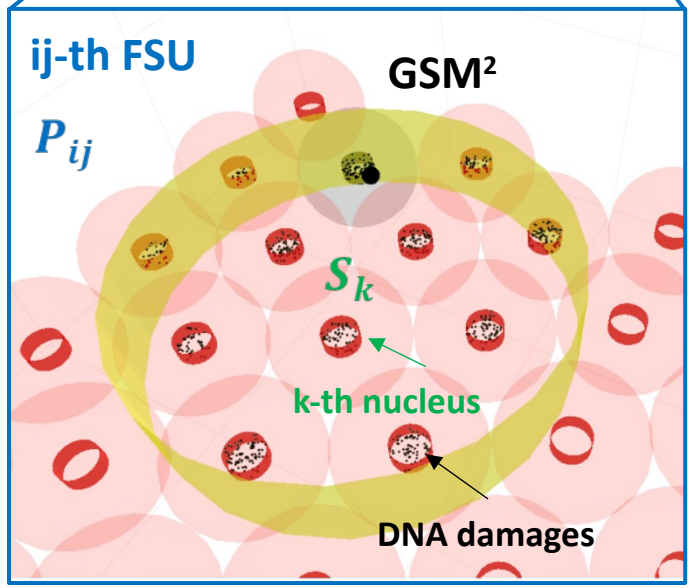
General model validation

WP1.2

Model extension to clinic

WP1.3



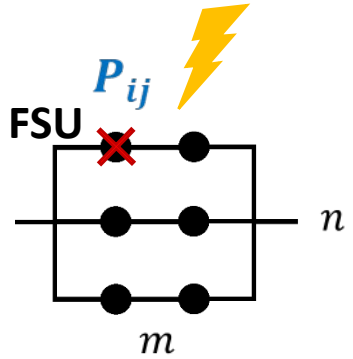


FSU response

Surviving probability of a single cell

$$P_{ij}\left(\mathrm{LET}_{ij}, D_{ij}, [\mathrm{O}_2]_{ij}\right) = \prod_{k=1}^{N_c}\left(1 - S_{ij}^{k}\left(z_n^k, \mathrm{LET}^k, X^k, Y^k \middle| \mathrm{LET}_{ij}, D_{ij}, [\mathrm{O}_2]_{ij}\right)\right)$$

Probability that all cells in the FSU die



Volume effect parameter

$$s := \frac{m}{m \cdot n} = \frac{1}{n}$$

- 0 Parallel organ
- 1 Serial organ

Organ response

$$\mathrm{NTCP} = \prod_{j=1}^{n}\left[1 - \prod_{i=1}^{m}\left(1 - P_{ij}\left(\mathrm{LET}_{ij}, D_{ij}, [\mathrm{O}_2]_{ij}\right)\right)\right]$$

Irradiation scheme of the skin tissue experiment

Irradiation of 3D skin tissue

WP2.1

Skin model development

WP2.2

Skin model calibration

WP2.3

Goal: investigation of the **biological and chemical processes** induced by **X-ray** and **proton beam** radiation at different irradiation conditions

Irradiation type	Total dose	Fractionation	Dose per fraction
- X-ray - Proton (entrance channel) - Proton (mid Spread-Out Bragg Peak (SOBP))	- 5.0 Gy - 15.0 Gy	- Single fraction (1f) - 5 fractions (5f)	- 1.0 Gy (5f) - 3.0 Gy (5f) - 5.0 Gy (1f) - 15.0 Gy (1f)

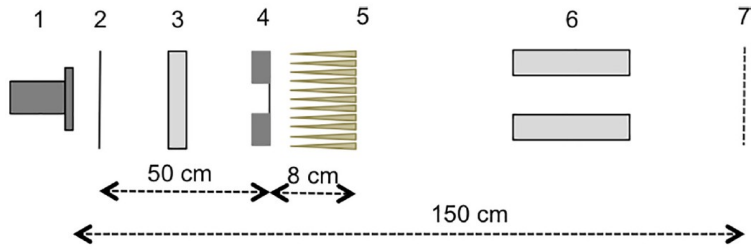
Orthovoltage X-ray @ TIFPA: 195 kVp, 10 mA

Experimental proton beam for radiobiology @ TPTC:
148 MeV nominal energy, dual-ring setup

Radiation quality characterization with microdosimetry:
Benchmark between experiments and MC TOPAS simulations

M. Missiaggia et al., Int. J. Radiat. Oncol. Biol. Phys. (2023)

A. Bianchi et al., Radat. Phys. Chem. (2023)



1 Beam pipe
2 First Foil
3 Monitor Chamber
4 Dual Ring
5 Range Modulator
6 Collimation (optional)
7 Target Position

F. Tommasino et al., Phys. Medica (2019)